

**Object Oriented Programming
(CS1004)**
Date: April 6th 2024

Course Instructor

Ahmed Hamza , Shazia Haque

Sessional-II

Total Time: 1 Hour

Total Marks: 30

Total Questions: 03

Semester: SP-2024

Campus: Lahore

Student Name

Roll No

Section

Signature

CLO2: Design a program using object-oriented programming paradigm.

Question No. 1 (CLO 2, PLO 1, C5)

Marks: 10

Generate the output of the following code

<pre> class M { public: virtual void MyMemory() { cout<<"I forget"<<endl; } void Disk() { cout<<"Space"<<endl; } void Erased() { cout<<"For good"<<endl; } void ThisExam() { Erased(); MyMemory(); } ~M(){} }; </pre>	<pre> class N:public M { public: void MyMemory() { cout<<"Gone"<<endl; } void Disk() { cout<<"Slipped"<<endl; } virtual void Erased() { cout<<"Rubbed out"<<endl; } }; void main() { M * m1=new N; m1->MyMemory(); m1->Disk(); m1->ThisExam(); cout<<endl; M m2=(new N); m2.MyMemory(); m2.Disk(); m2.ThisExam(); } </pre>
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Handwritten output for the first object (m1):
Gone
Space
For good
Gone

Handwritten output for the second object (m2):
I forget
Space
For good
I forget

Question No. 2 (CLO 2, PLO 1, C5)

Marks: 10

Design the solution to the following problem using C++

You are supposed to write the software needed by a company to maintain its computer systems. Each computer in the company has a unique serial_no, it also has a fixed RAM, ROM, processor and manufacturer. A computer can either be a laptop. Each laptop has a screen_size and weight. A computer has many softwares installed on it. Each software has a unique license_no, it also has a cost, is_system_software, manufacturer and version.

- Identify the classes needed. [2 marks]
- Identify any inheritance relationships [1 marks]
- Identify any composition relationships [1 marks]
- In which order are the constructors called when a Laptop is initialized [2 marks]
- Give the prototype of the classes such that in an array of pointers of Computers if we call print method laptop info is printed differently if the pointer is referencing a laptop. [4 marks]

Question No. 3 (CLO 2, PLO 1, C5)

Marks: 10

Consider the following prototype of course and section classes and write code in C++ to answer the following questions.

```
class Course
{
    int courseNumber, creditHours;

public:
    void setCourse (int courseNum,int creditHrs);//already implemented
};

class Section
{
    int secNumber;
    Course c;
public:
    void setSection (int courseNum,int creditHrs,int section);// to be implemented
};
```

- What sort of relationship exists between the classes Course and Section? [1 marks]
- Assuming that the setCourse member function of the Course class is already implemented give an implementation of the setSection method of the Section class. [4 marks]
- Write a main program that declares an array of 7 objects of type section and set their values as follows: [5 marks]

Course number: 117 for all sections.

Credit hours: 3 for all sections.

Section number: give each section a unique number from 1-7.